

Power and Spares

File: power.kicad_sch

Master Control

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PCI Bus

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ATA Interface

File: IDE.kicad_sch

Agnus Multiadapter

File: Agnus-Multi.kicad_sch

HID and User Ports

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Chip RAM Controller

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Audio

File: audio.kicad_sch

PCI Bridge

File: pci-bridge.kicad_sch

Video

File: video.kicad_sch

Motorola Bus Logic

File: motorola_bus_logic.kicad_sch

Chip Memory

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Clocks, ROM, USB

Serial

File: serial.kicad_sch

Floppy and Parallel

File: floppy.kicad_sch

File: clocks.kicad_sch
PCI Bus Logic

PCI Bus Logic

File: pci_bus_logic.kicad_sch

100 = PCI BUS
200 = CHIPSET
300 = VIDEO
500 = AUDIO
600 = POWER
700 = CHIPSET RAM
800 = VOLTAGE SHIFTER
900 = ATA

BUDDY

DESIGNED WITH KICAD REV 8

Sheet: /
File: AmigaPCI Mainboard.kicad_sch

Title: AMIGA PCI

Size: USLedger Date: 2026-06-06

Rev: 7.0

Size: 05Ledge1	
KiCad E.D.A. 8.0.9	

Id: 1/17



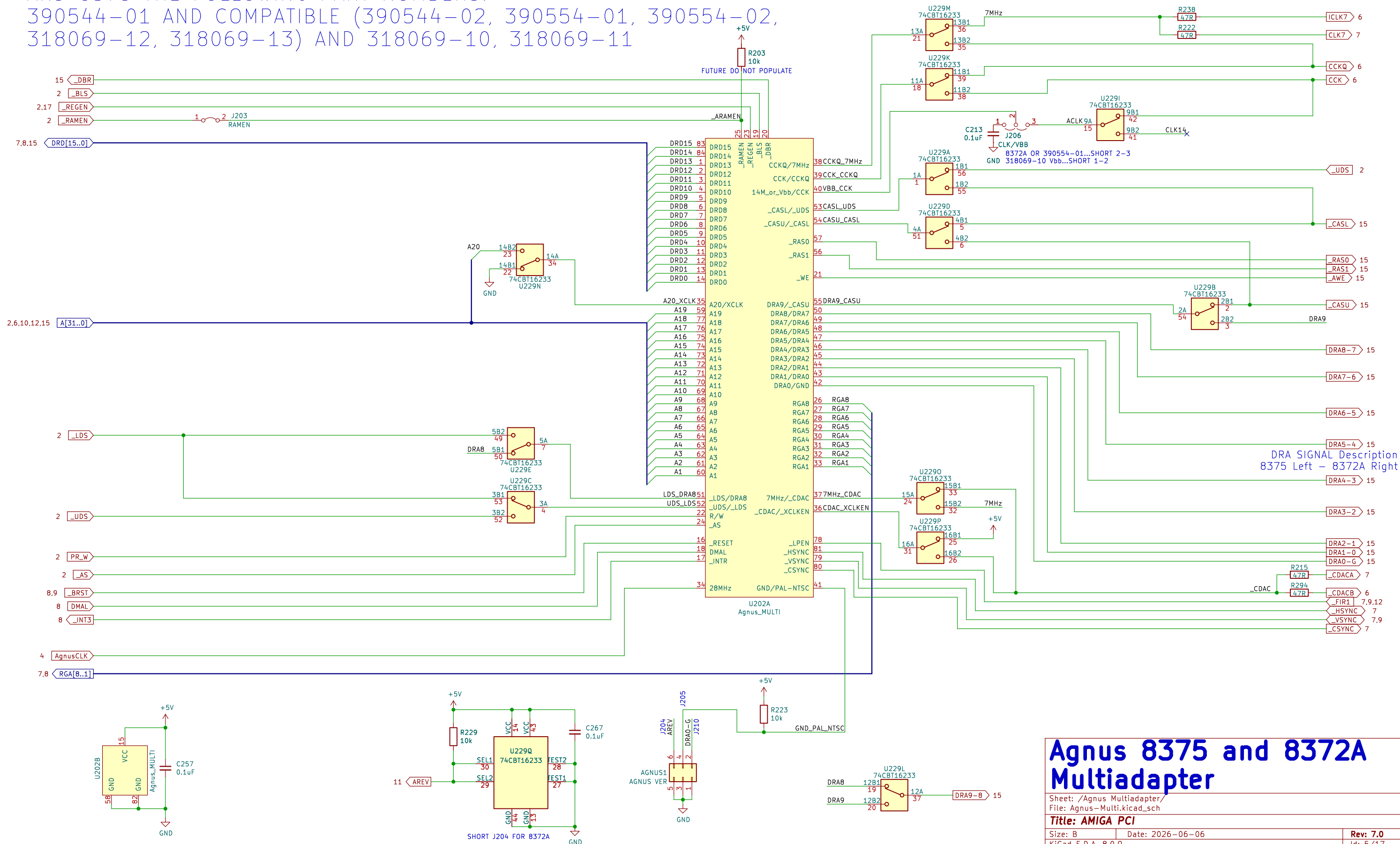
Id: 3/17

MASTER CONTROL

SUPPORTS 8372A (ALL PART NUMBERS)
AND 8375 THE FOLLOWING PART NUMBERS:
390544-01 AND COMPATIBLE (390544-02, 390554-01, 390554-02,
318069-12, 318069-13) AND 318069-10, 318069-11

JUMPERS MUST BE CONFIGURED BY AGNUS PART NUMBER

The diagram shows a circuit for configuring jumpers based on the Agnus part number. It features a yellow component labeled U229M (74CBT16233). A +5V supply is connected to pin 13A (pin 21). Pin 13B1 (pin 36) is connected to a 7MHz clock source. Pin 13B2 (pin 35) is connected to ground. The output of the component is connected to two red jumper boxes labeled R228 and R222, both with a value of 47R. These jumpers are connected to the ICLK7 and CLK7 pins, which are labeled 6 and 7 respectively.

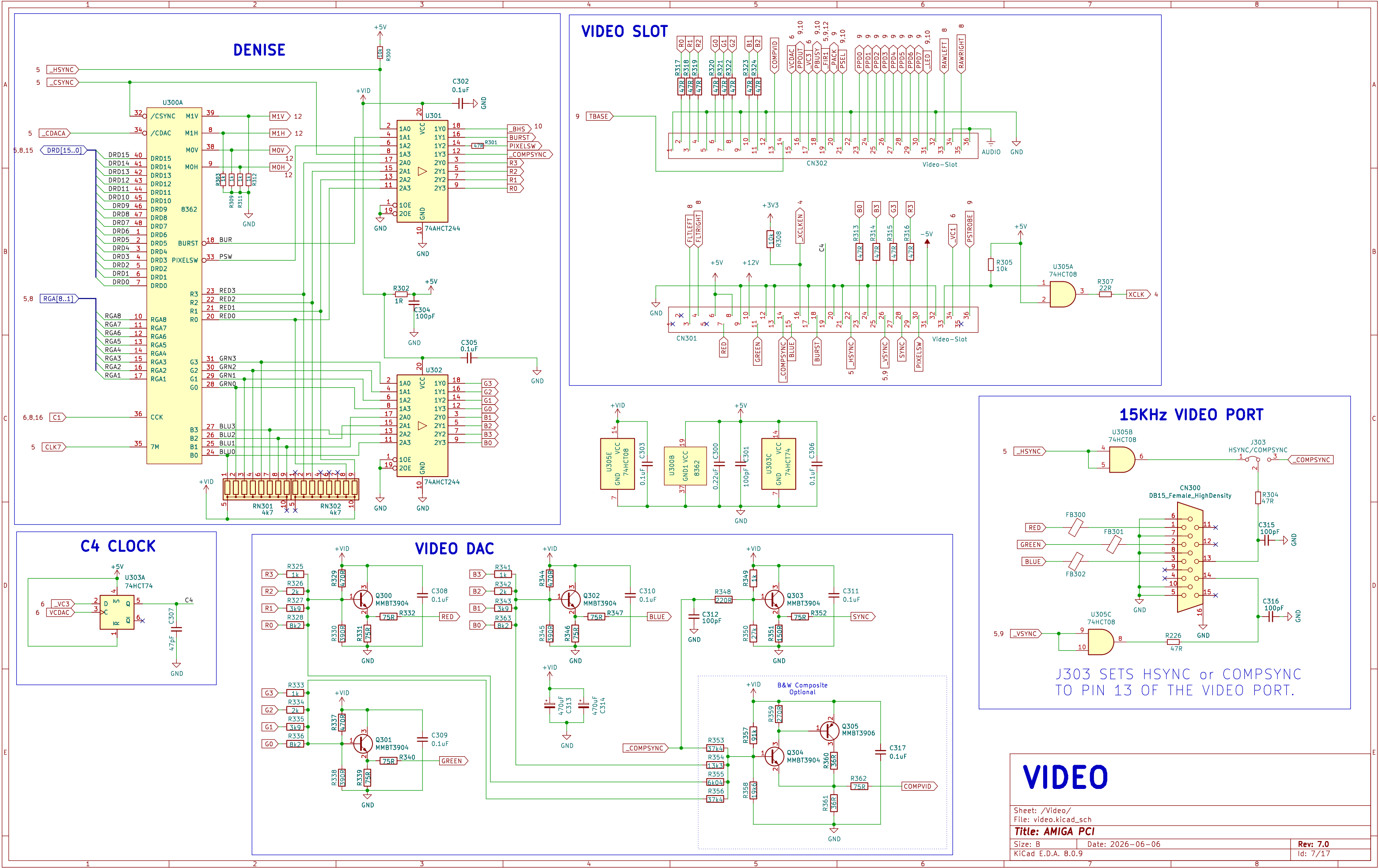


Agnus 8375 and 8372A Multiadapter

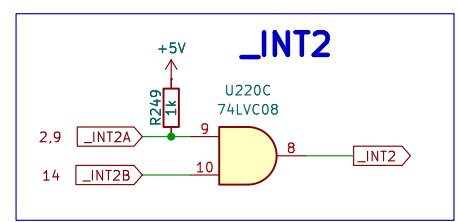
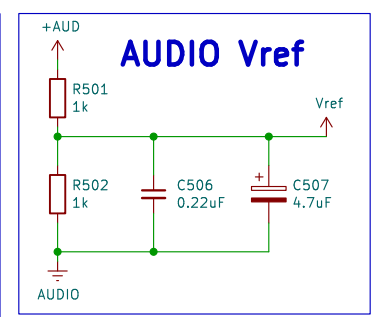
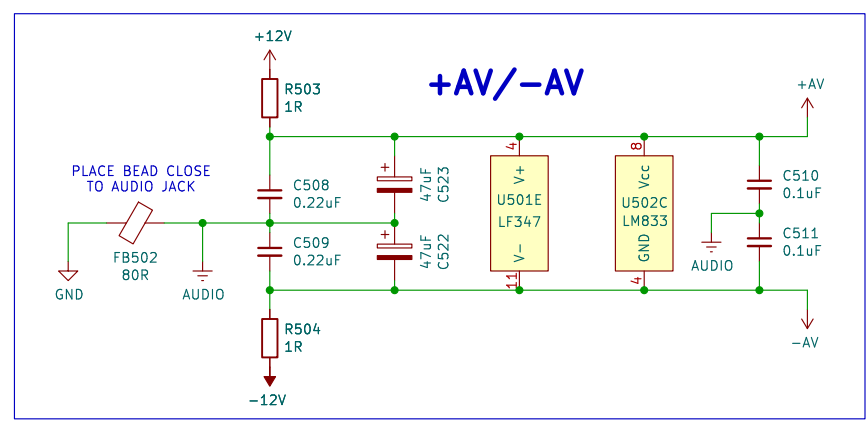
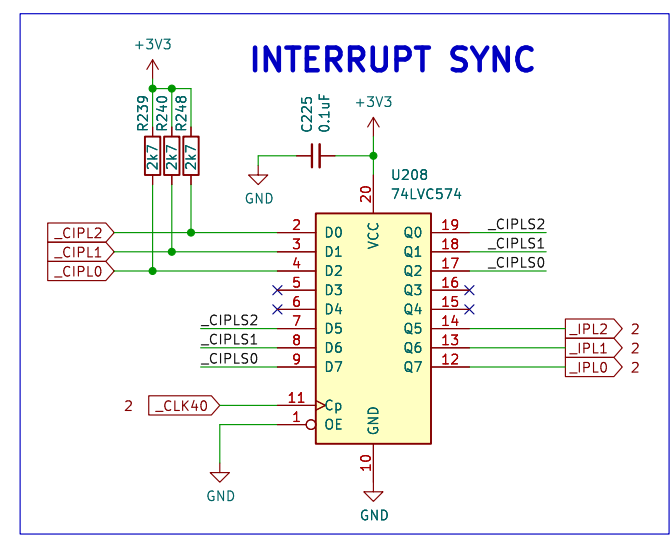
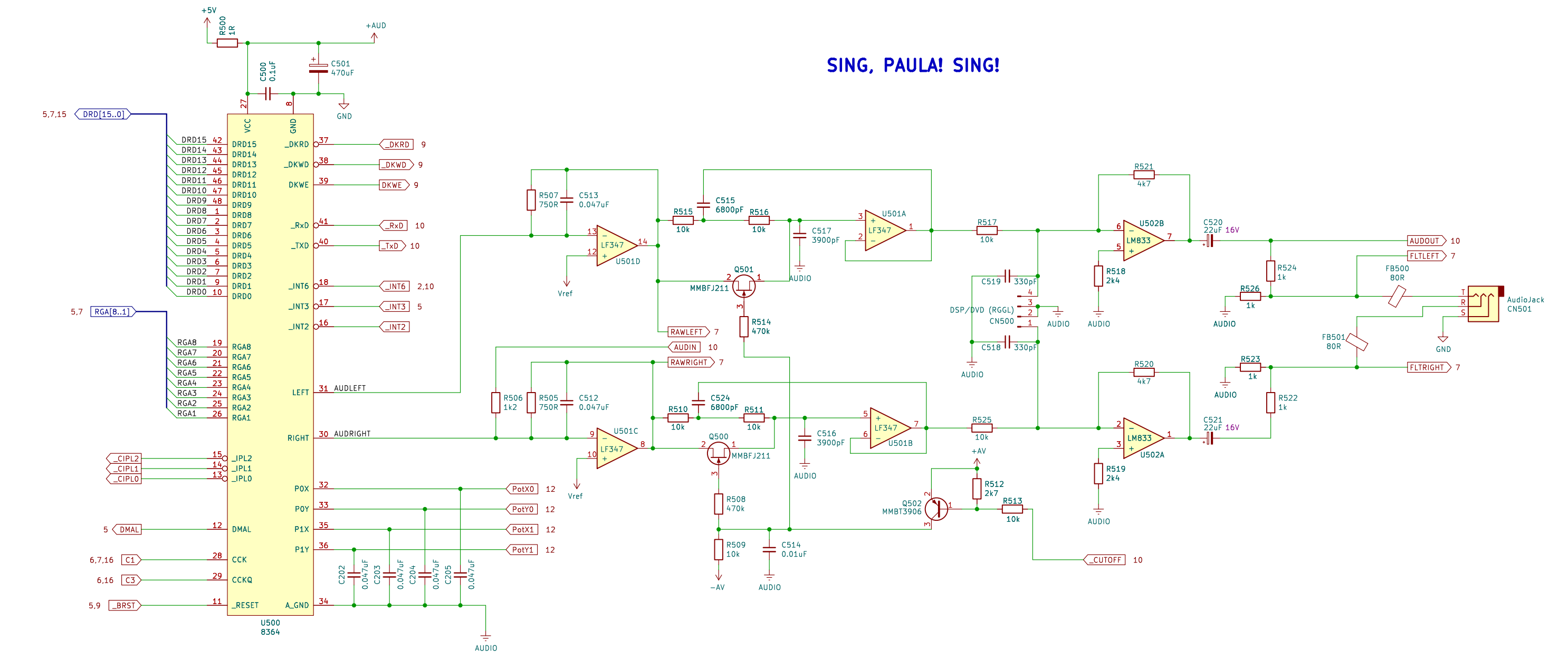
Sheet: /Agnus Multiadapter/
File: Agnus-Multi.kicad_sch

Title: AMIGA PCI

Size: B	Date: 2026-06-06	Rev:
KiCad E.D.A. 8.0.9		Id: 5

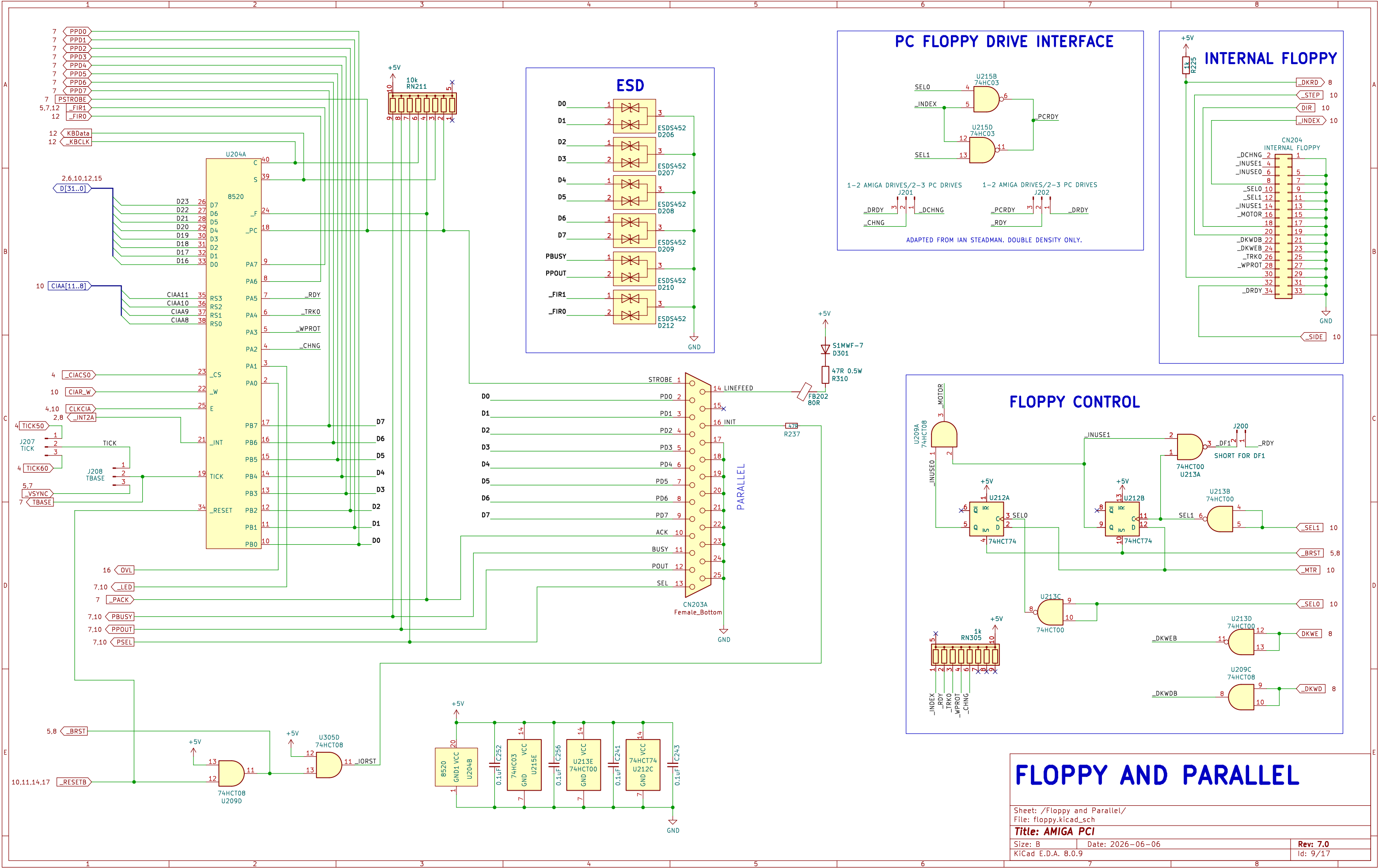


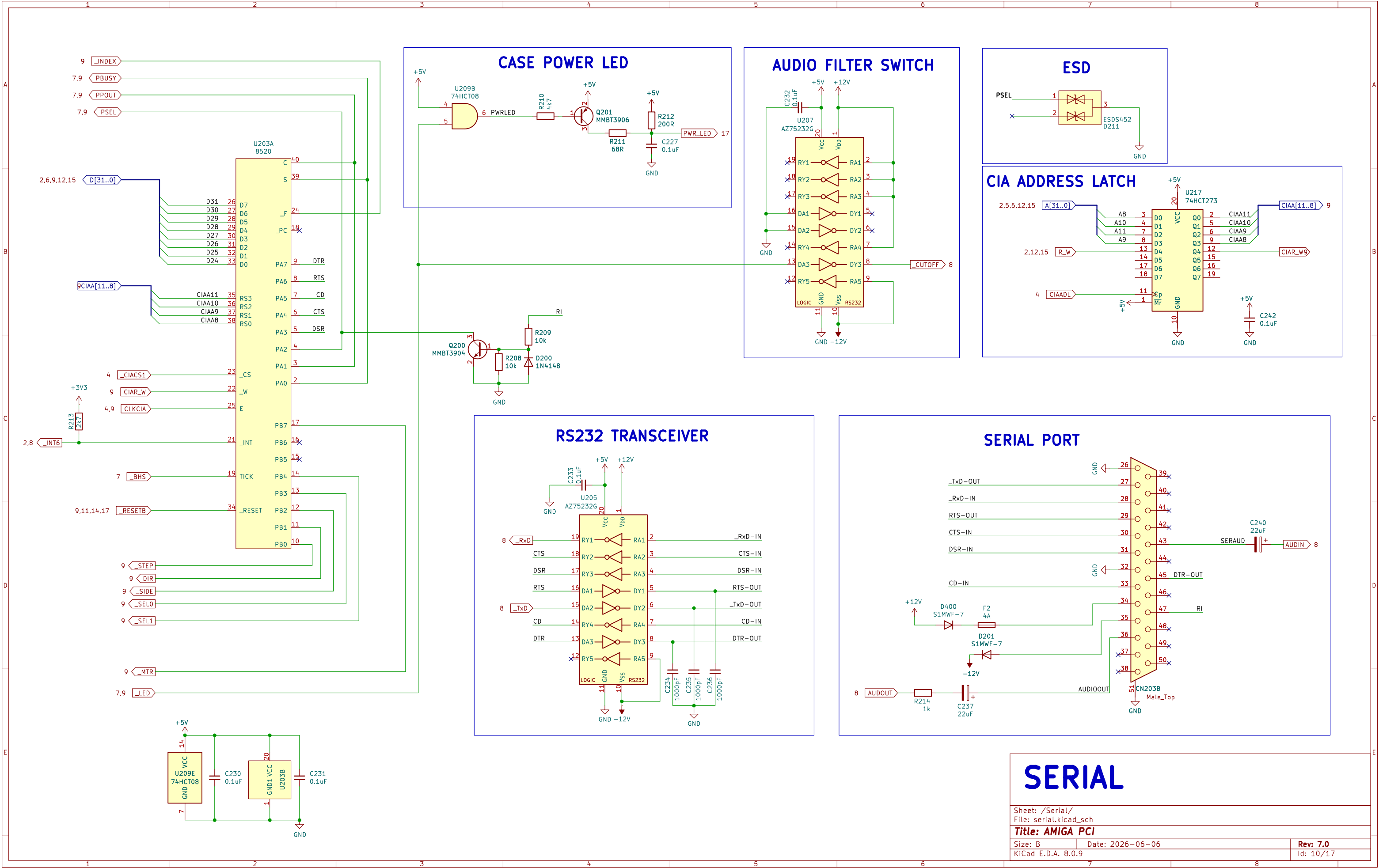
SING, PAULA! SING!



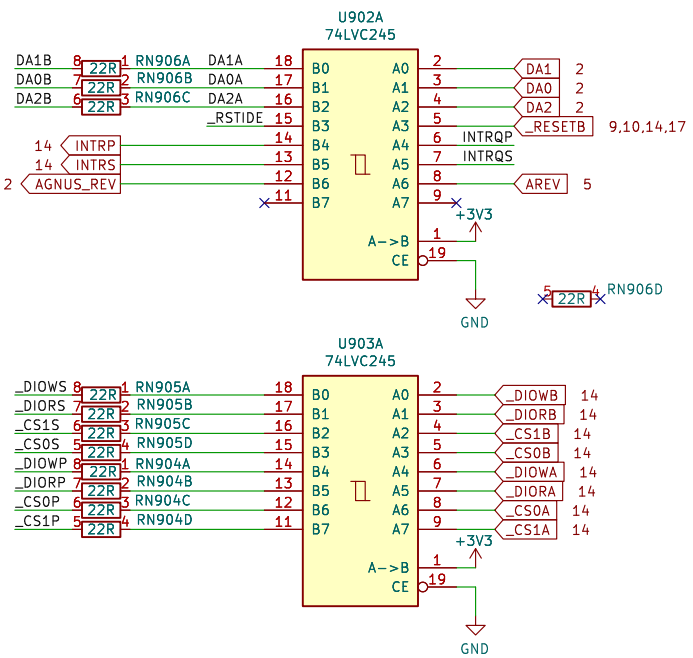
PAULA/AUDIO

Sheet: /Audio/ File: audio.kicad_sch		
Title: AMIGA PCI		
Size: B	Date: 2026-06-06	Rev: 7.0
KiCad E.D.A. 8.0.9		Id: 8/17

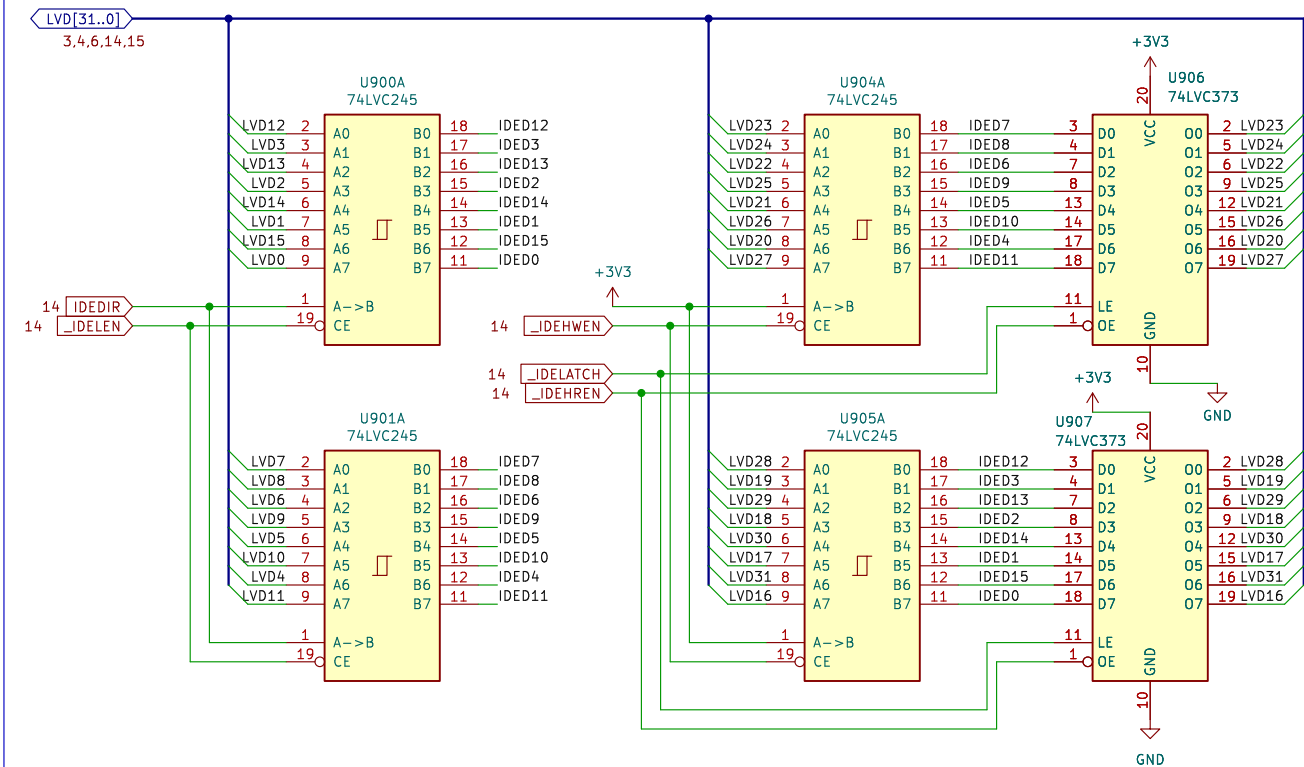




BUFFERS AND TERMINATION



DATA BUFFERS



ATA INTERFACE

Sheet: /ATA Interface/
File: IDE.kicad_sch

Title: AMIGA PCI

Size: B

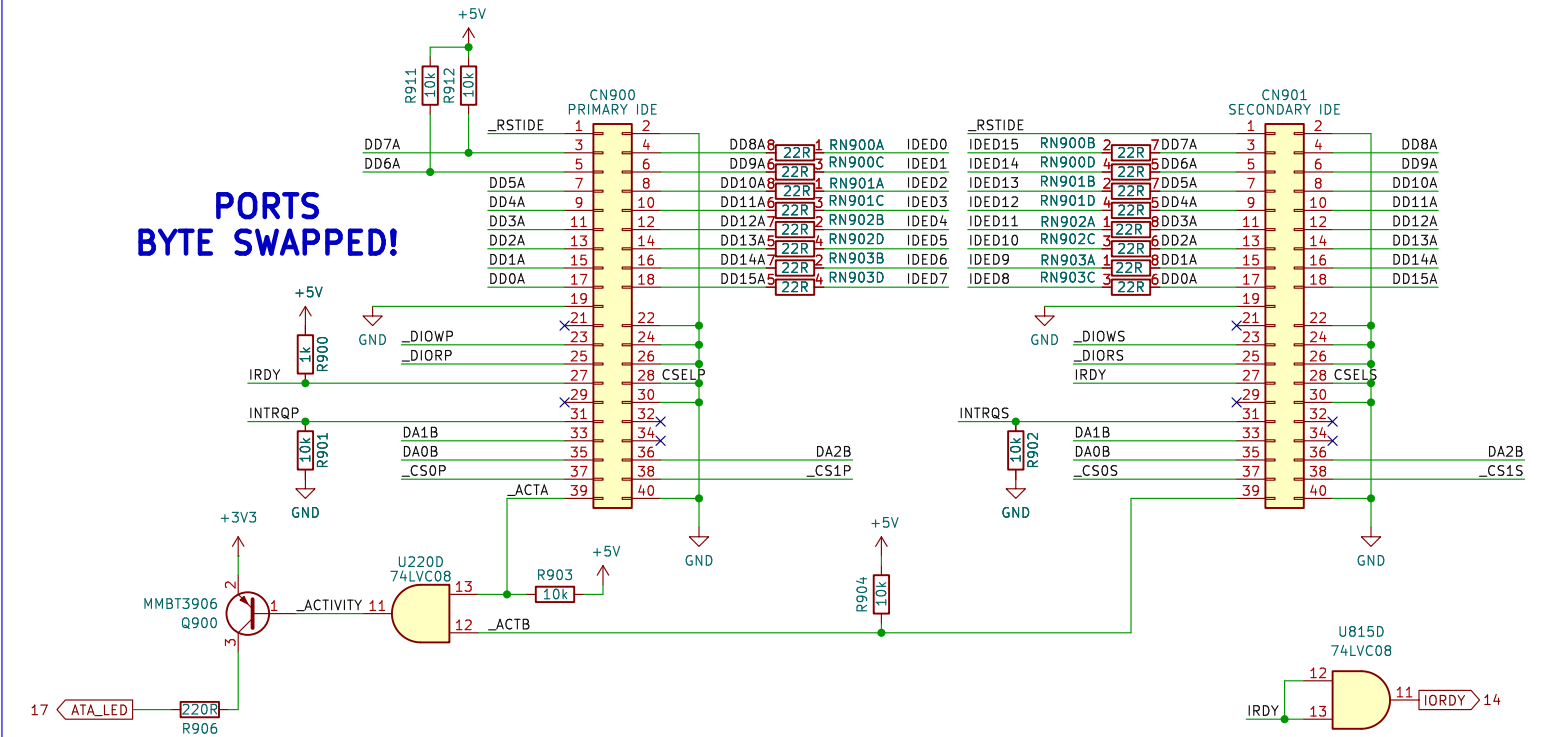
Date: 2026-06-06

Rev: 7.0

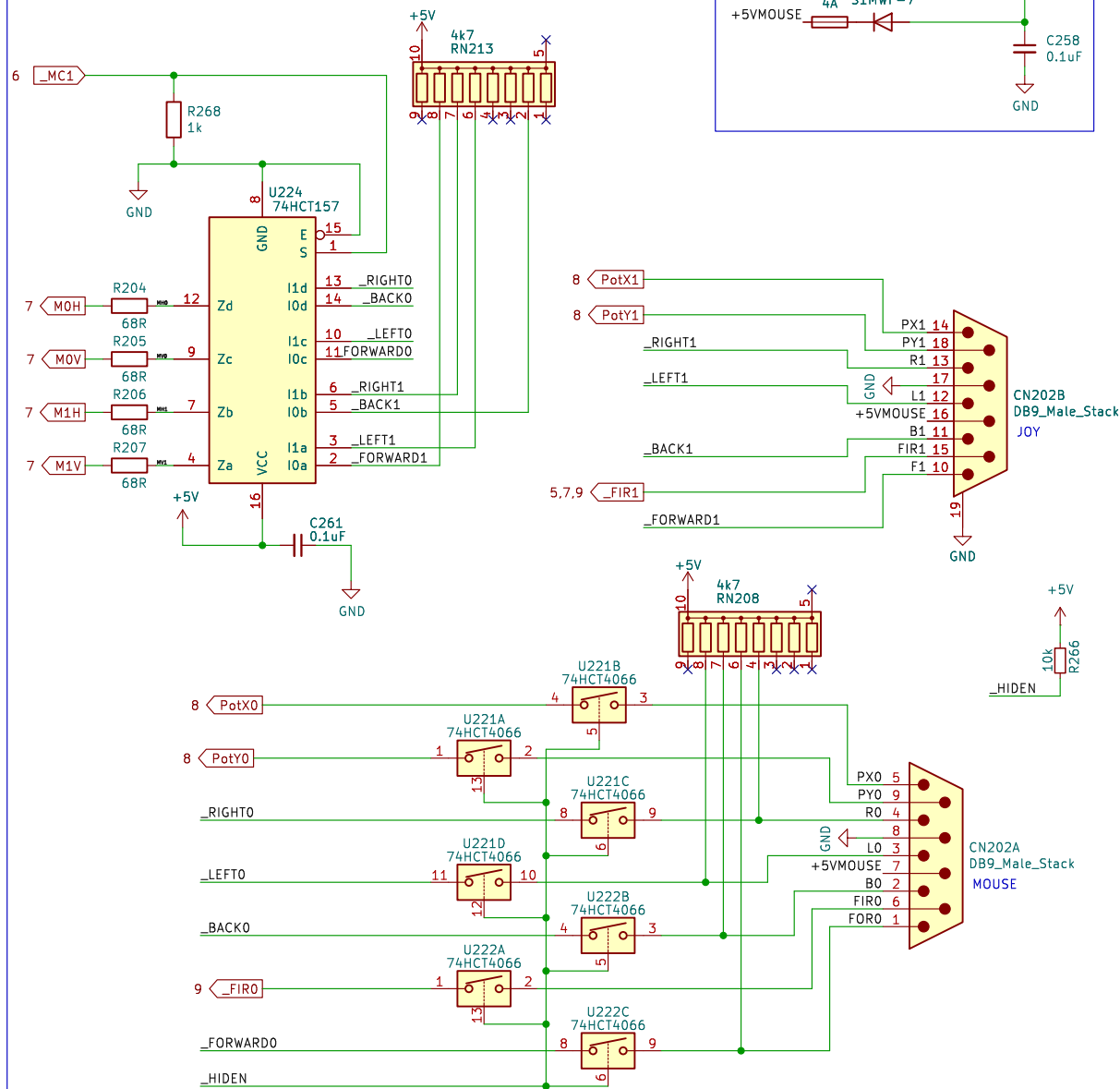
KiCad E.D.A. 8.0.9

Id: 11/17

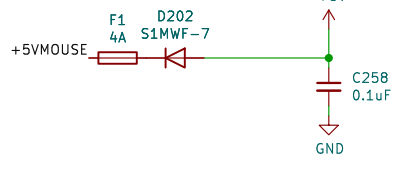
PORTS BYTE SWAPPED!



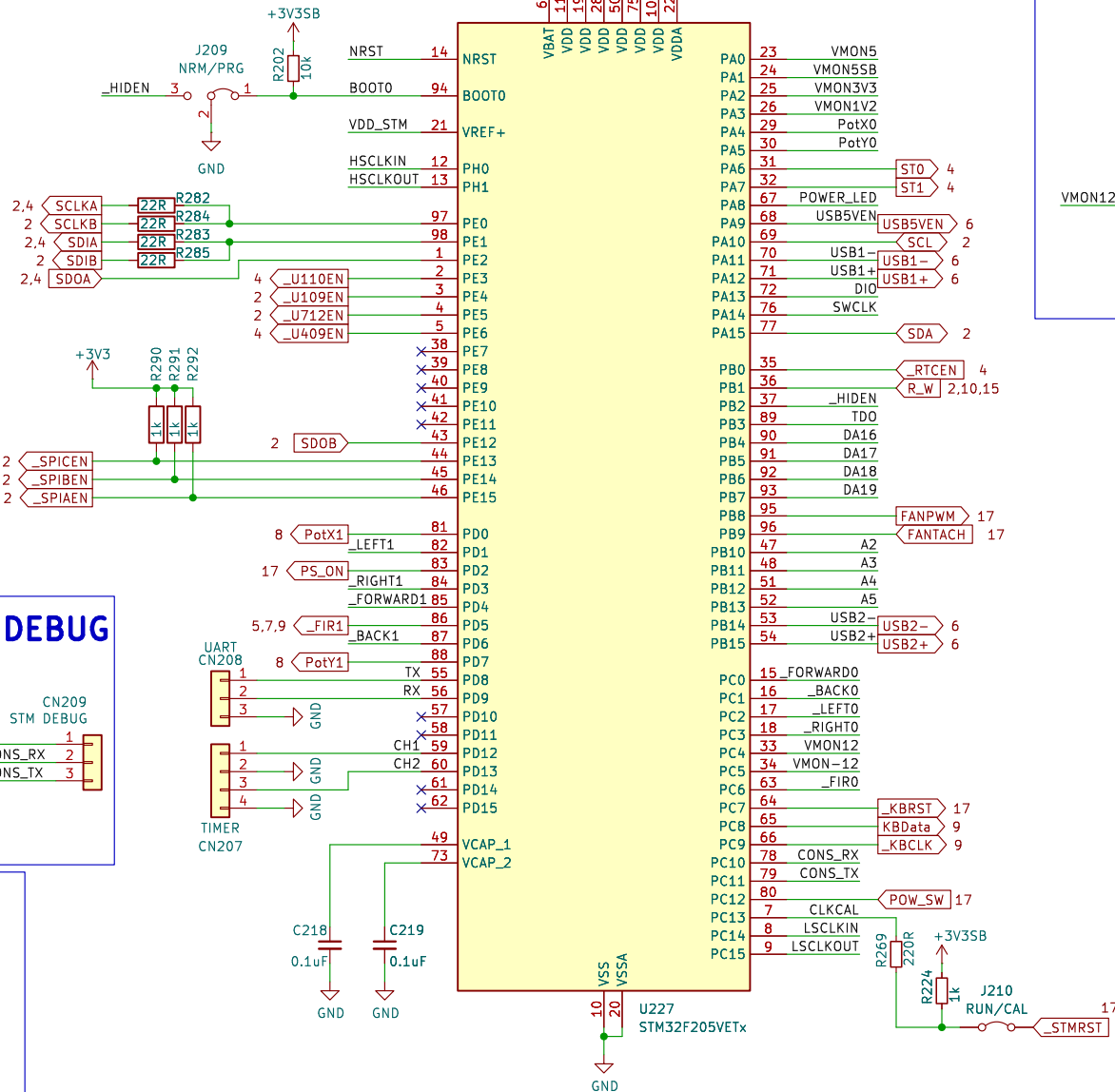
LEGACY MOUSE/JOYSTICK



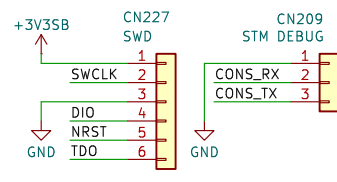
MOUSE/JOY POWER



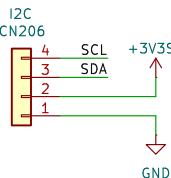
HID INTERFACE, REAL TIME CLOCK, AND OTHER AWESOME THINGS.



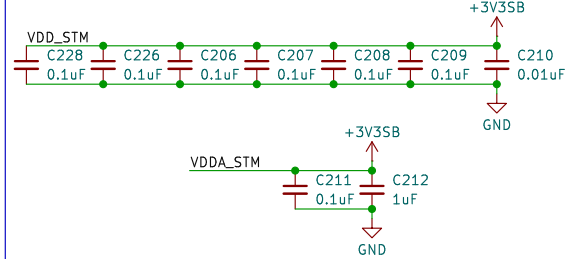
STM PRGM & DEBUG



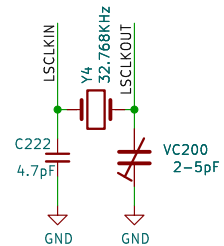
I2C



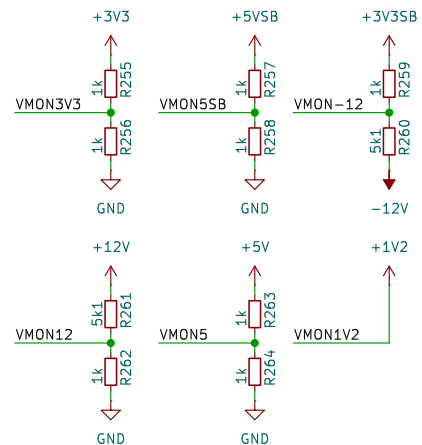
STM POWER



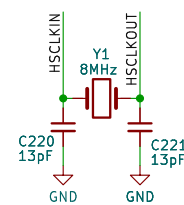
CALENDAR OSC



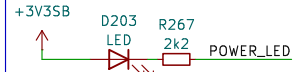
VOLTAGE DIVIDERS



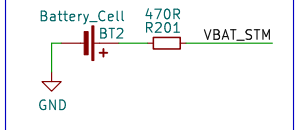
STM OSC



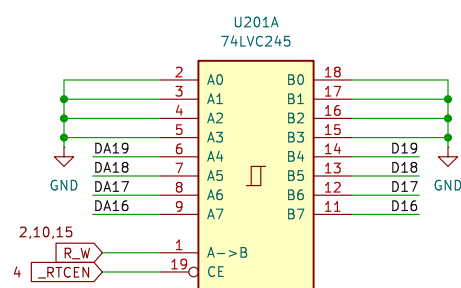
STM STATUS LED



BACKUP BAT



RTC DATA BUFFERS



HID and USER PORTS

Sheet: /HID and User Ports/
File: hid.kicad_sch

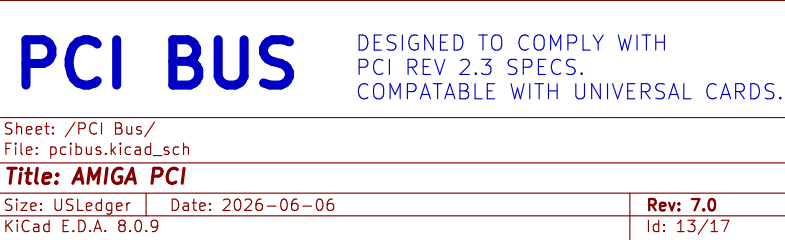
Title: AMIGA PCI

Size: B Date: 2026-06-06

KiCad E.D.A. 8.0.9

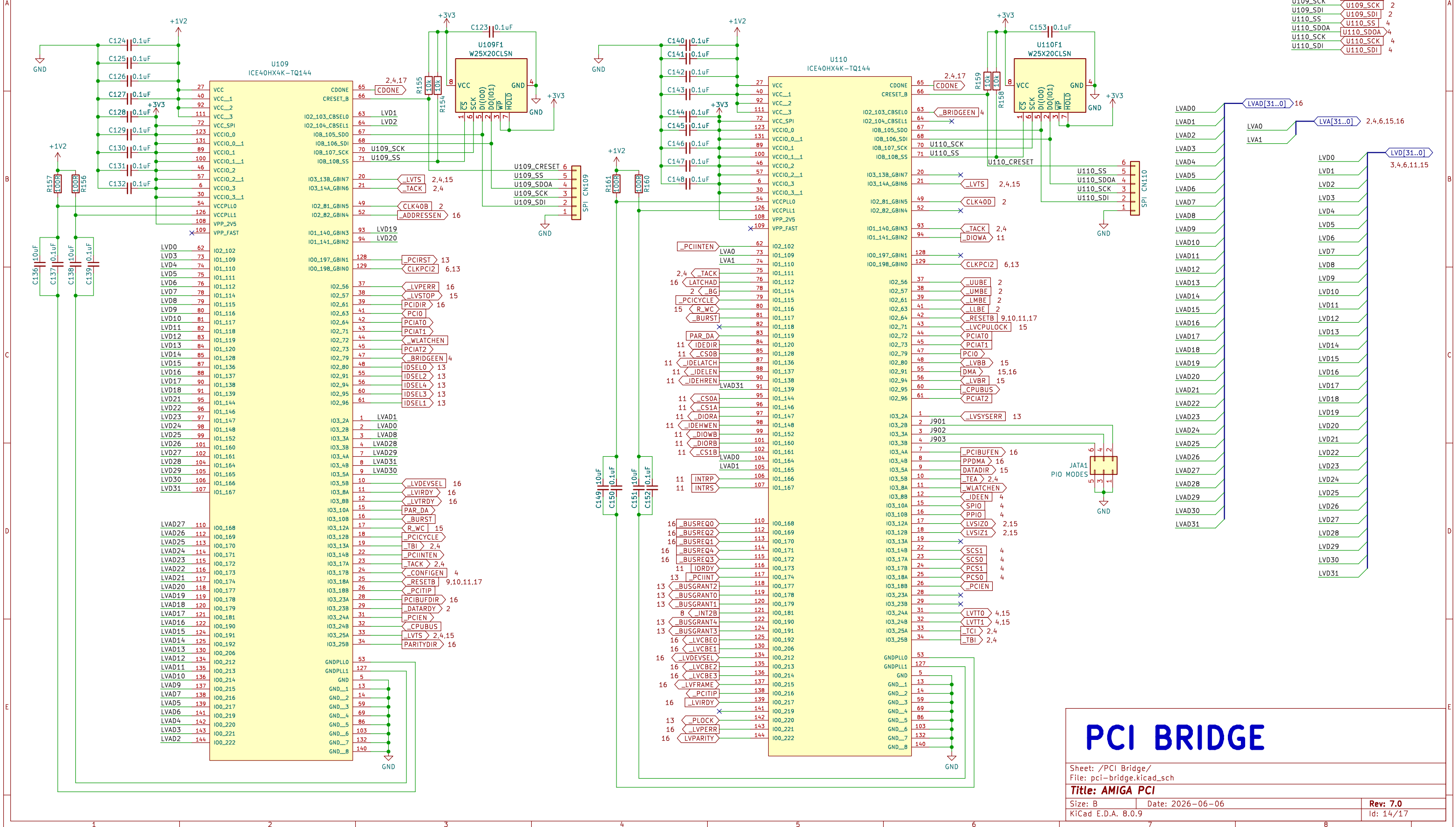
Rev: 7.0

Id: 12/17

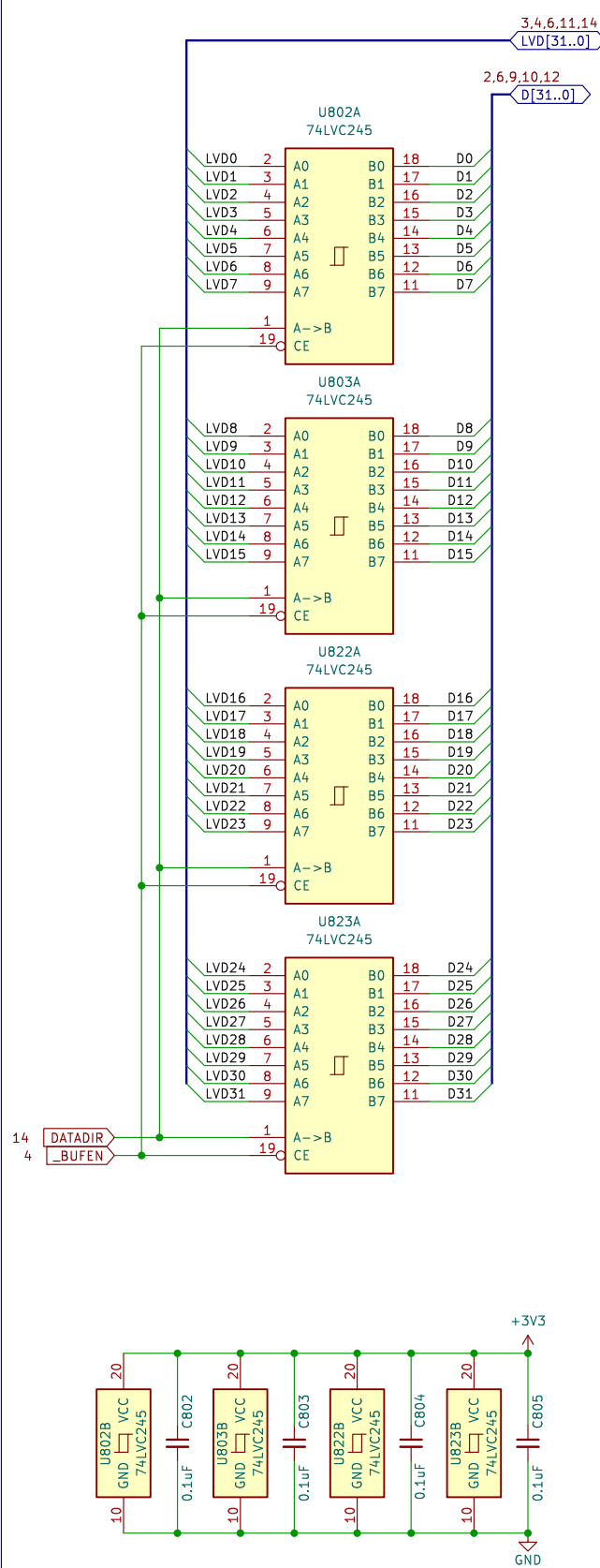


U109: BUFFERS AD/D/A BUSSES, AD PARITY CALC, BRIDGE REGISTERS

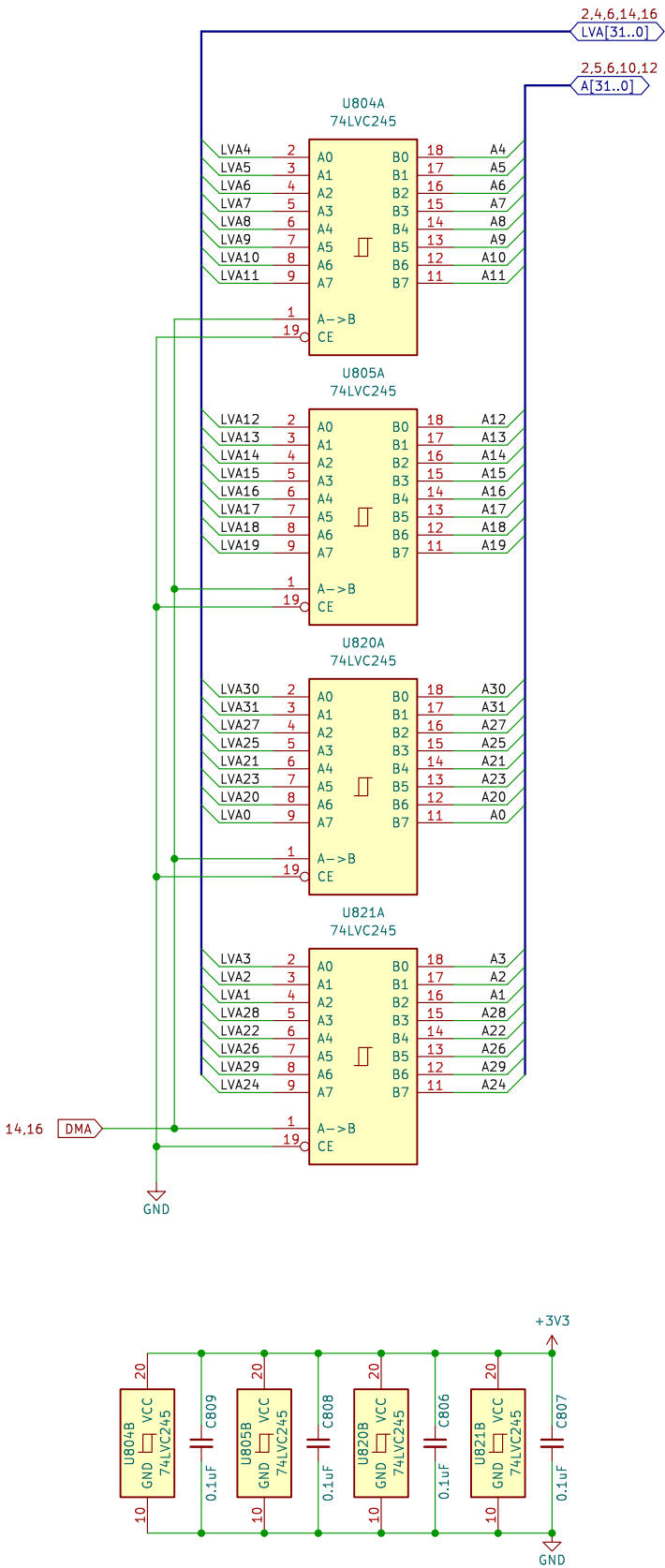
U110: BUS ARBITOR, INTERRUPT HANDLING, PARITY ERROR DETECTION, ATA CONTROLLER



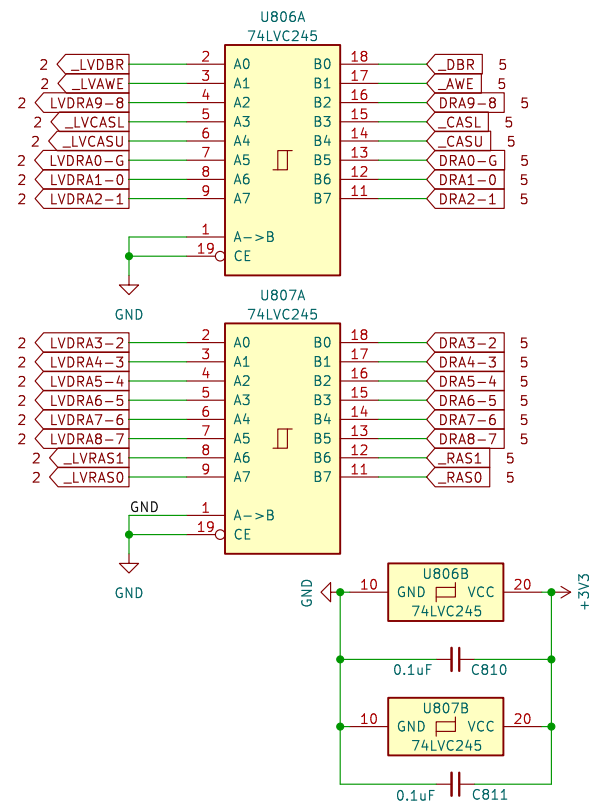
LOW VOLTAGE DATA BUFFERS



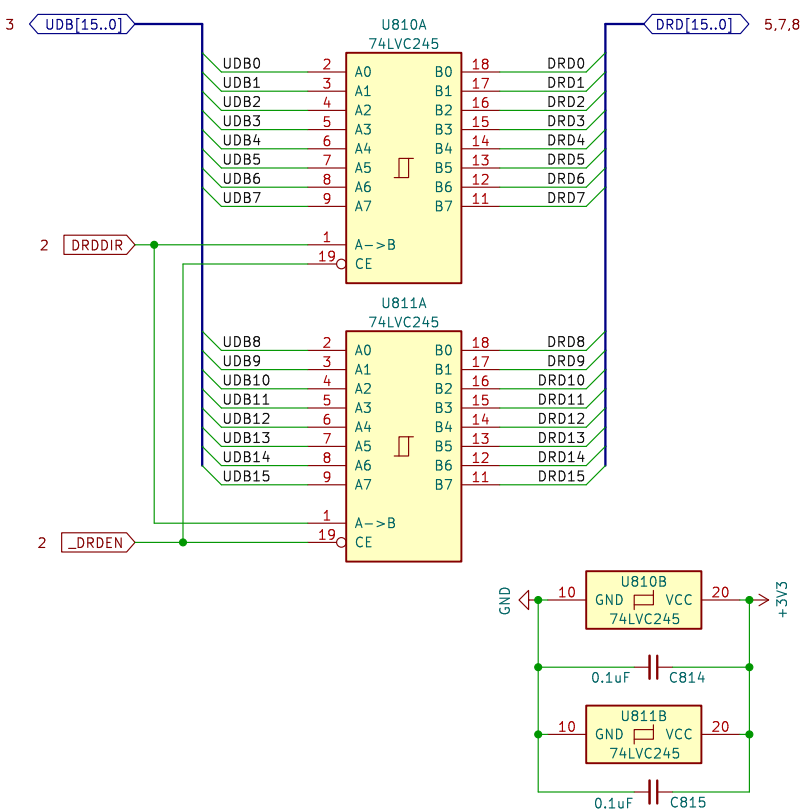
LOW VOLTAGE ADDRESS BUFFERS



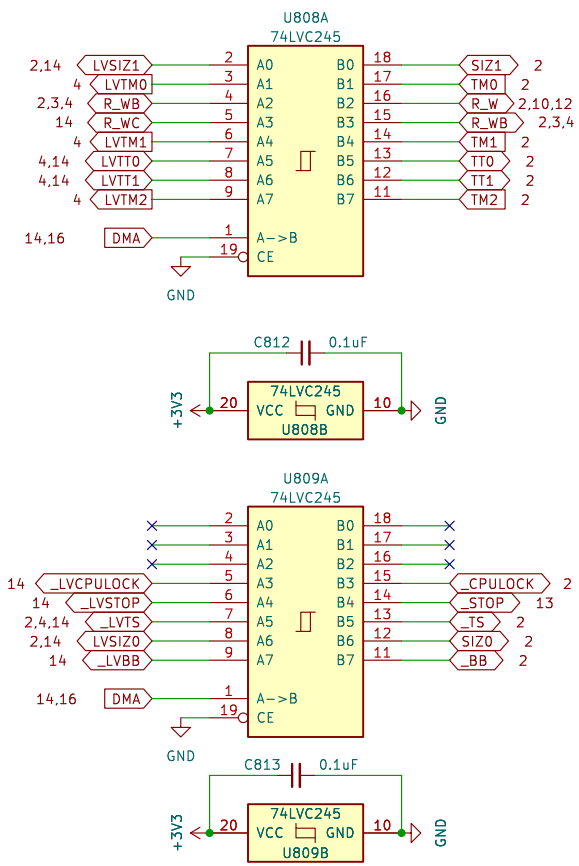
LOW VOLTAGE AGNUS SIGNALS



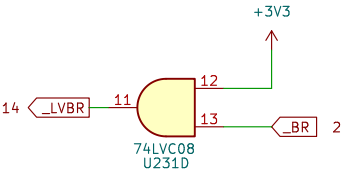
CHIPSET RAM DATA BUS BUFFER



BIDIRECTIONAL MC68040

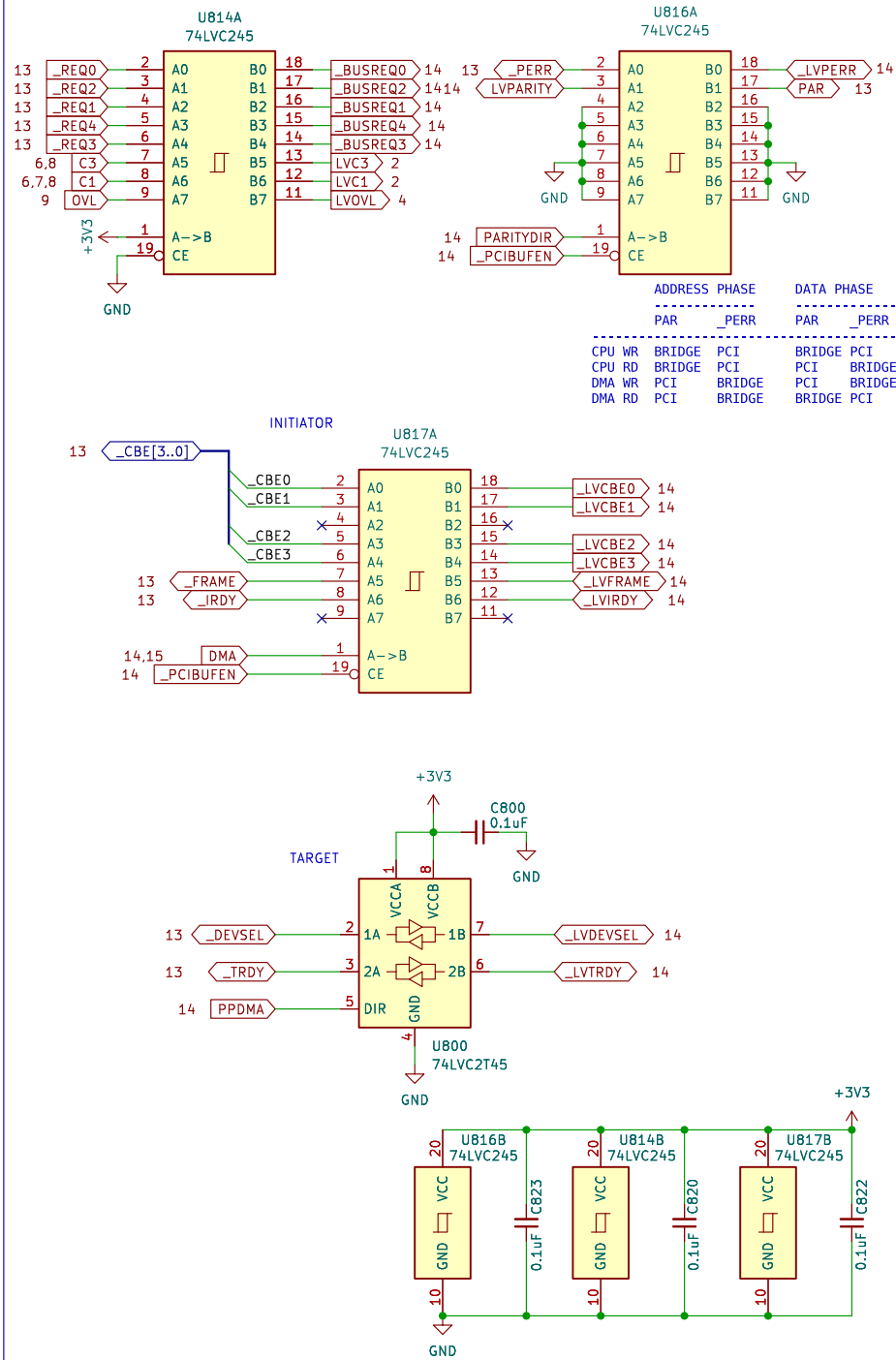


BUS REQUEST

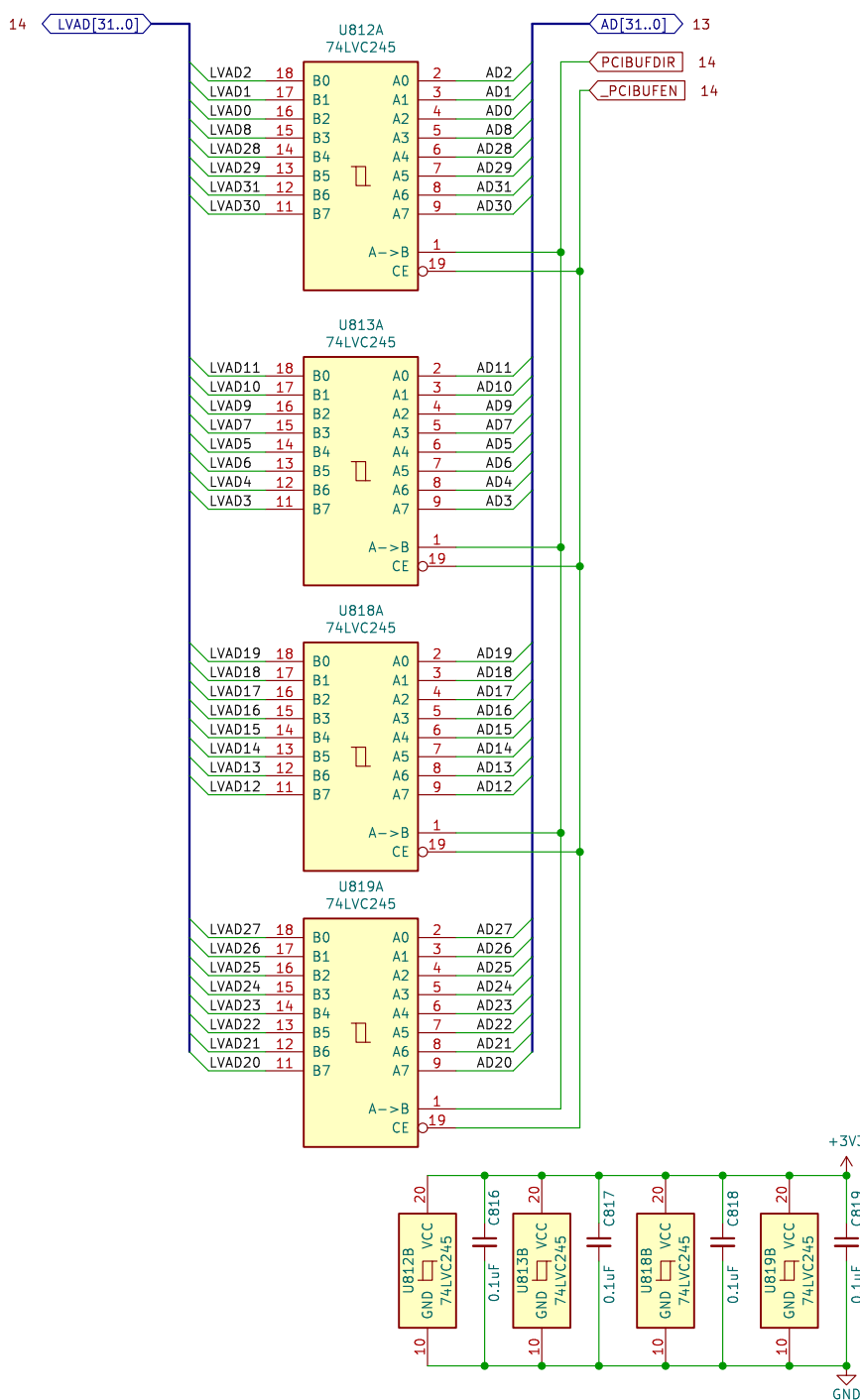


MOTOROLA BUS LOGIC

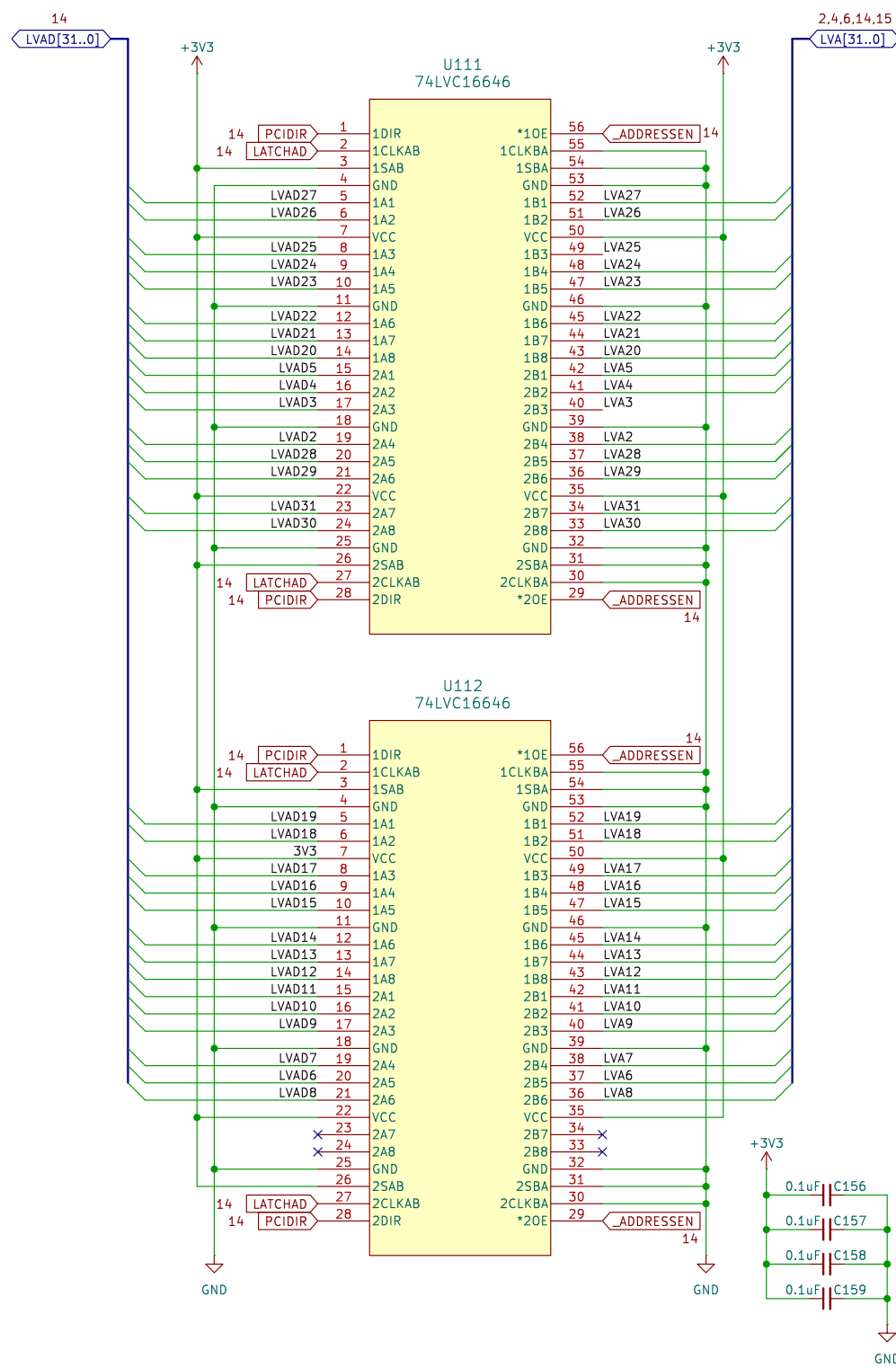
PCI BUFFERS



AD BUS LEVEL SHIFT



PCI ADDRESS LATCH



PCI BUS LOGIC

Sheet: /PCI Bus Logic/

File: pci_bus_logic.kicad_sch

Title:

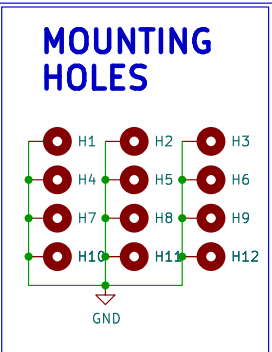
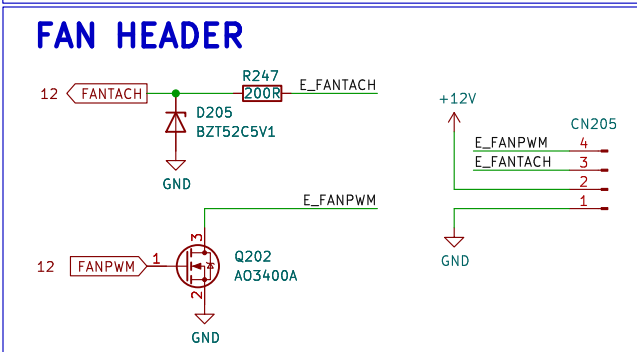
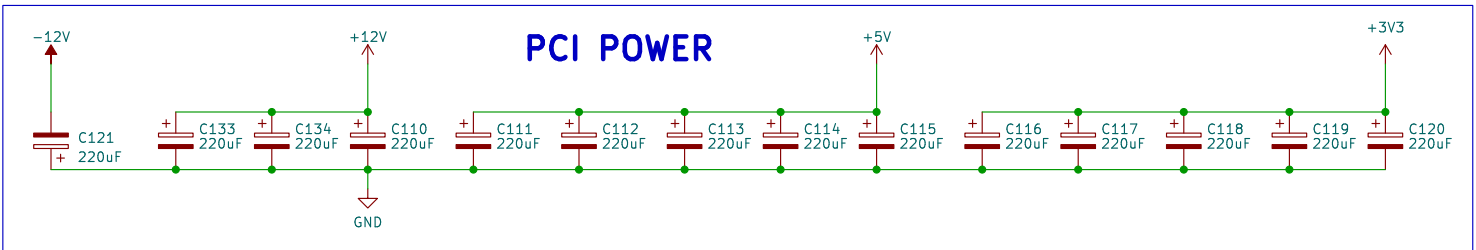
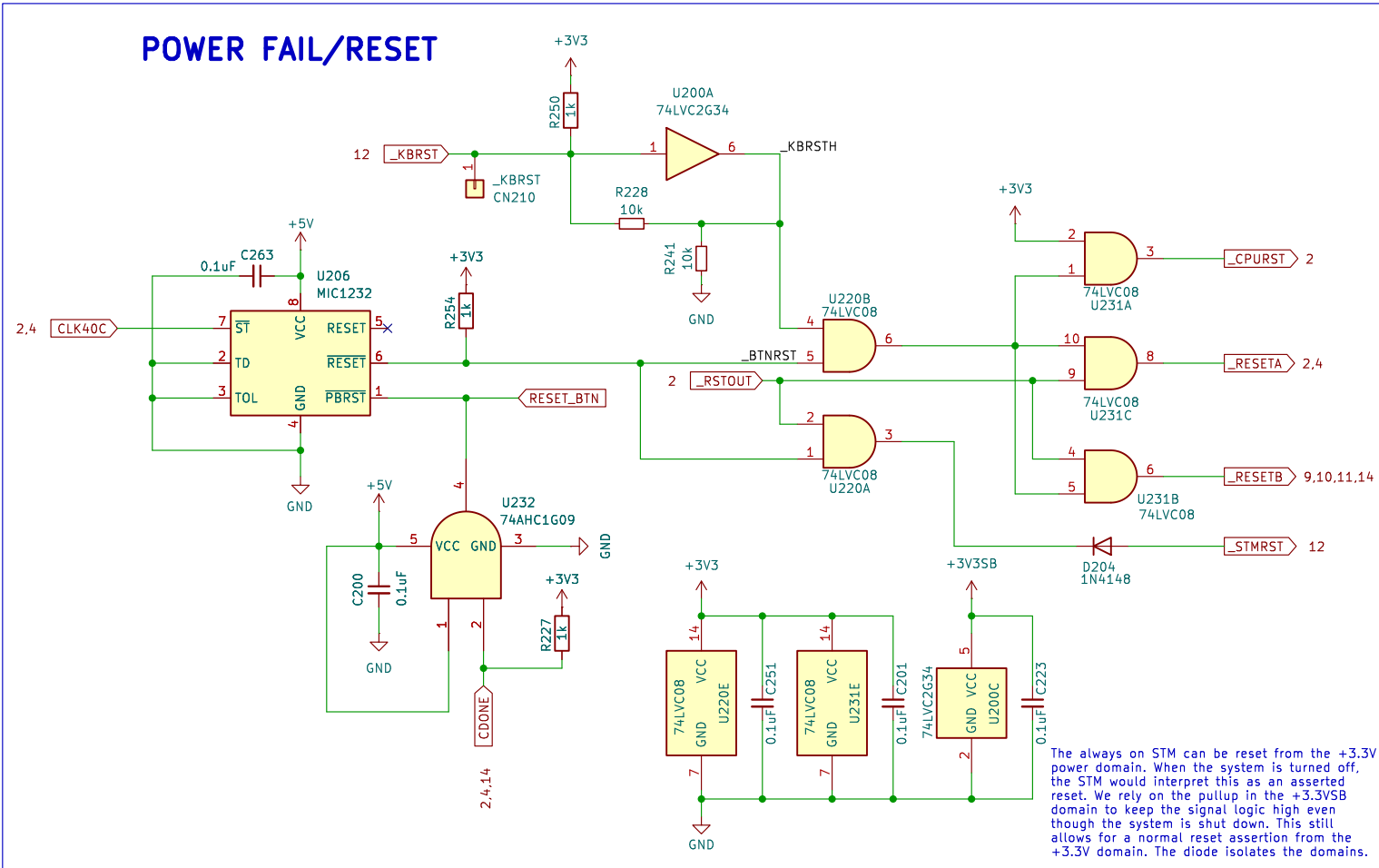
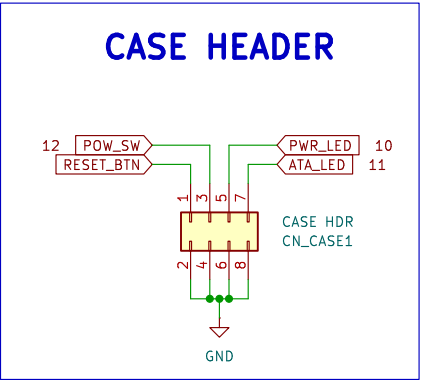
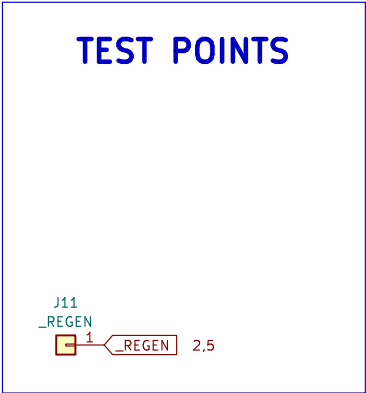
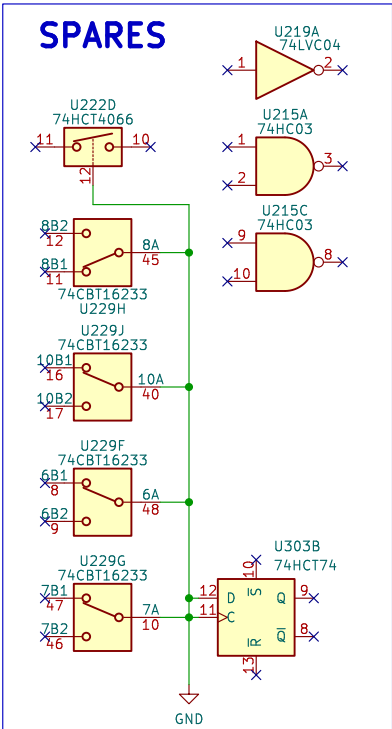
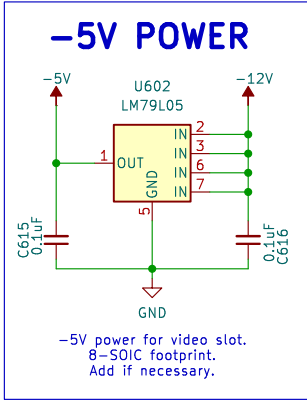
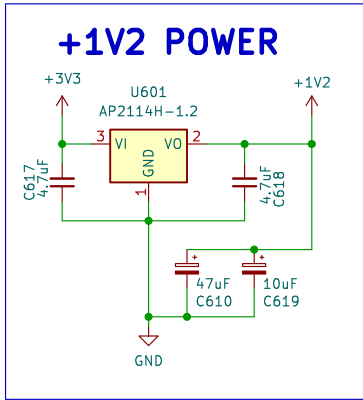
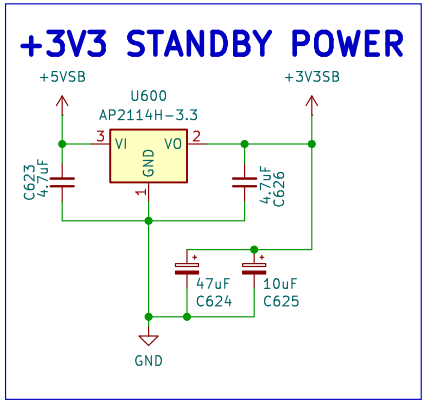
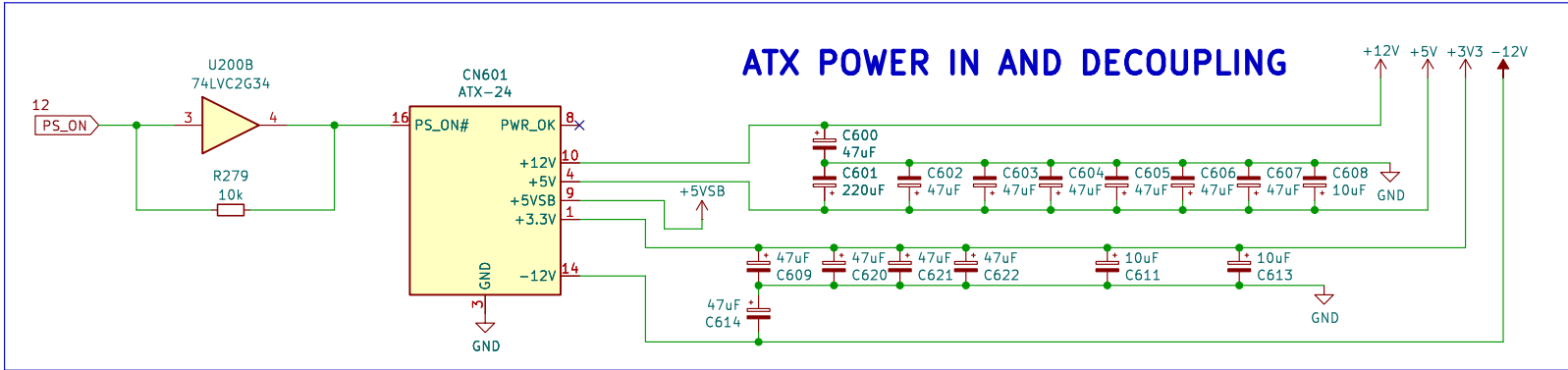
Size: B

Date: 2026-06-06

Rev: 7.0

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Id: 16/17



POWER/SPARES

Sheet: /Power and Spares/

File: power.kicad_sch

Title: AMIGA PCI

Size: B

Date: 2026-06-06

Rev: 7.0

KiCad E.D.A. 8.0.9

Id: 17/17